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|  <b>Marwadi</b><br>University | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                  |
| <b>Subject: Digital Signal and Image Processing (01CT1513)</b>                                                 | <b>Aim :</b><br><b>Audio conclusion</b>                                                                                    |                                  |
| <b>Open handed problem -2</b>                                                                                  | <b>Date:17-07-2026</b>                                                                                                     | <b>Enrolment No: 92400133131</b> |

```

Code 1:- import librosa
import soundfile as sf
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import convolve

audio, sr = librosa.load(r"D:\sem-5\dsip\Haan_Karde.mp3", sr=None, mono=True)

kernel = np.array([1, 0, 1, 0, 1], dtype=np.float32)

convoluted = convolve(audio, kernel, mode="same")

convoluted /= np.max(np.abs(convoluted))

sf.write("output_convoluted.wav", convoluted, sr)

print("Saved as output_convoluted.wav")

audio /= np.max(np.abs(audio))

plt.figure(figsize=(12, 6))

plt.subplot(2, 1, 1)
plt.plot(audio)
plt.title("Original Audio")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.subplot(2, 1, 2)
plt.plot(convoluted)
plt.title("Convoluted Audio")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.tight_layout()
plt.show()

```

|                                                                                                             |                                                                                                                            |                                  |
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Output :-

